4.3 Rotations

Geometry 4.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- rotate a figure 90°, 180°, or 270° about the origin.
- perform a composition with a rotation.
- tell whether a figure has rotational symmetry.

Vocabulary · fill in the missing word

Rotation — a _____ in which a figure is turned about a fixed point, the center of rotation

Angle of rotation — the angle _____ which a figure is turned (counterclockwise unless stated)

Rotational symmetry — a _____ has rotational symmetry when a rotation of 180° or less maps it onto itself

Part 1 · Discovery: turn it

Plot P(3, 1). Rotate it counterclockwise about the origin with tracing paper.

Rotation	90°	180°	270°
Image of P(3, 1)			

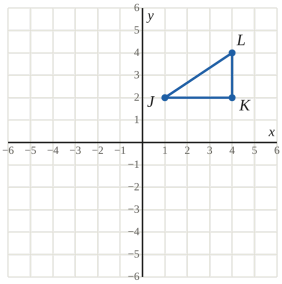
What patterns do you see in the coordinates?

Part 2 · Notes and examples (fill in each note, then work the example)

Rotations about the origin (counterclockwise): 90°: (x, y) → _____; 180°: (x, y) → _____; 270°: (x, y) → (y, −x).

Example 1

Rotate △JKL 90° about the origin. Give the image's vertices.



Example 2

a. Rotate (−3, 5) 180° about the origin.

b. Rotate (2, −6) 270° about the origin.

Example 3

Rotate (2, 3) 90° about the origin, then reflect the image in the x-axis.

Example 4

Which rotation about the origin maps (4, 1) to (−1, 4)?

Rotational symmetry: a figure has it when a rotation of _____ or less (about its center) maps it onto itself.

Example 5

Does the figure have rotational symmetry? If so, give the angles.

a. a regular pentagon

b. a parallelogram

c. an isosceles trapezoid

Part 3 · Partner practice

Partner 1

Rotate (5, −2) 90° about the origin.

Partner 2

Rotate (−4, −1) 180° and (separately) 270° about the origin.